Mathias Labonté

mlabonte@ucdavis.edu
mlabonte@lbl.gov
(+1) 530-219-3417

ORCID ◇ Google Scholar ◇ INSPIRE-HEP

Department of Physics
One Shields Avenue
Davis, CA 95616

Lawrence Berkeley
National Laboratory
1 Cyclotron Road
Berkeley, CA 94720

EDUCATION

University of California, Davis — 2022 – Present
Doctor of Philosophy, Nuclear Physics — *Expected June 2027*
Advisor: Daniel Cebra

Niels Bohr Institute, University of Copenhagen — 2020 – 2022
Master of Science, Particle Physics
Advisor: You Zhou

University of British Columbia — 2016 – 2020
Bachelor of Science *(Hons)*, Medical Physics

AFFILIATIONS

Lawrence Berkeley National Laboratory — 2023 – Present
Affiliate, Relativistic Nuclear Collisions Group, Nuclear Science Division — *Berkeley, CA*

Brookhaven National Laboratory — 2023 – Present
User, Relativistic Heavy Ion Collider (STAR Collaboration) — *Upton, NY*

University of California, Davis — 2022 – Present
Graduate Student Researcher, Nuclear Physics Group — *Davis, CA*

SCIENTIFIC HIGHLIGHTS

- **Heavy-ion collisions:** Measured light-hadron production, baryon stopping, and the thermodynamic properties of matter at high baryon density in STAR fixed-target Au+Au collisions; searched for the QCD critical point with net-proton fluctuations.
- **Photonuclear physics and the Electron-Ion Collider:** Measured the size of the baryon number distribution inside the proton, using historical $\gamma+p$ data; compared nuclear target species to inform the choice of beam for the EIC early science program.

- **Machine learning:** Determined collision centrality with neural networks; member of the UC Davis team on a DOE Genesis Mission award (2026–) applying AI to learn the nuclear equation of state.
- **Instrumentation:** Evaluated the performance of the STAR Endcap Time-of-Flight detector; diagnosed time-projection chamber misalignment in fixed-target track reconstruction.

AWARDS AND FELLOWSHIPS

STAR Early Career Award March 2025

For "Extensive investigations into track reconstruction in the fixed target data that ultimately led to the discovery of TPC misalignment issues."

Brady Fellowship October 2024

Departmental fellowship for research performance and community outreach.

PUBLICATIONS

As a STAR author since 2023, I am an author on 50+ publications. A comprehensive list can be found here: https://www.star.bnl.gov/central/publications/. Publications in which I served in a principal capacity are listed below, as well as authorship outside of the STAR collaboration.

Separating coherent and incoherent photoproduction at an Electron-Ion Collider: comparing lead, gold and silver beams
Spencer R. Klein, **Mathias C. Labonté**
Preprint (2026) [arXiv:2609.30168]

Light hadron production measurements with Au+Au collisions from $\sqrt{s_{\rm NN}}$ = 3.2–4.5 GeV with STAR
Mathias C. Labonté (for the STAR Collaboration)
J. Subatomic Part. Cosmol. ***6*** *(2026) 100484 — conference proceedings* [arXiv:2607.22945, Journal]

Imaging baryon number density within the proton
Spencer R. Klein, **Mathias C. Labonté**, Gerald A. Miller, Zachary Sweger, Ramona Vogt
Preprint (2026) [arXiv:2603.03730]

Performance of the Endcap Time-of-Flight detector in the STAR beam-energy scan
Mathias C. Labonté, Daniel Cebra, Zachary Sweger, Geary Eppley, Frank Geurts, Yannick Söhngen, Norbert Herrmann, Esteban Rubio, Philipp Weidenkaff, Ingo Deppner, Pierre-Alain Loizeau, Jochen Frühauf, David Emschermann, Florian Seck, David Tlusty, Dongdong Hu, Yongjie Sun, Tonko Ljubicic, Yi Wang
Nucl. Instrum. Meth. A ***1090*** *(2026) 171612* [arXiv:2602.16828, Journal]

INVITED AND PLENARY TALKS

Talks given at internal collaboration meetings are marked as such.

25 Years of STAR Meeting December 2025
Brookhaven National Lab, NY

Invited Plenary: *"STAR's Contributions to Understanding the Phase Diagram"*

APS Division of Nuclear Physics (DNP) 2025 October 2025
Chicago, IL

Invited Talk: *"Search for the QCD Critical point"*

2025 RHIC and AGS Annual Users' Meeting June 2025
Brookhaven National Lab, NY

Invited Plenary: *"The Endcap Time-of-Flight Detector at STAR"*

Early Career Award Presentation May 2025
Brookhaven National Lab, NY

STAR collaboration meeting. **Plenary Flash Talk**: *"When STARs don't align"*

11th Workshop of the APS Topical Group on Hadronic Physics March 2025
Anaheim, CA

Invited Talk: *"Tagging coherent vector-meson production events at the Electron-Ion Collider"*

STAR eTOF Workshop December 2024
Brookhaven National Lab, NY

STAR collaboration meeting. **Invited Plenary:** *"Comparing FXT and Collider modes"*

STAR eTOF Workshop December 2024
Brookhaven National Lab, NY

STAR collaboration meeting. **Invited Plenary:** *"Extending STAR phase-space coverage"*

STAR Analysis Meeting September 2024
Online

STAR collaboration meeting. **Invited Plenary:** *"Fixed-Target QA and TPC Alignment"*

STAR/CBM eTOF Workshop February 2024
Darmstadt, Germany

STAR/CBM collaboration workshop. **Invited Talk:** *"eTOF and its Implications for Light-Flavour Spectra Measurements with STAR"*

CONTRIBUTED TALKS AND POSTERS

Presented below is a selection of contributed talks and posters given at various conferences and meetings, and is not a comprehensive list. As a member of a large experimental collaboration, internal conferences are common.

APS April Meeting 2026 April 2026
Denver, CO

Talk: *"The Endcap Time-of-Flight Detector at STAR"*

Strangeness in Quark Matter 2026 March 2026
Los Angeles, CA

Talk: *"Bulk property measurements from Au+Au collisions in STAR FXT"*

California EIC Consortium Meeting June 2025
San Luis Obispo, CA

Talk: *"Searching for baryon junctions with photo-nuclear collisions"*

Quark Matter 2025 March 2025
Frankfurt, Germany

Poster: *"Light-Flavour Hadron Production and Baryon Stopping at High Baryon Density"*

California EIC Consortium Meeting August 2024
Davis, CA

Talk: *"Incoherent Events with BeAGLE"*

American Physical Society April Meeting April 2024
Sacramento, CA

Talk: *"Towards $\pi^{\pm}$, $K^{\pm}$, $p/\bar{p}$ hadron production measurements in fixed target collisions with STAR"*

APS Division of Nuclear Physics (DNP) November 2023
Kona, HI

Talk: *"Light-Flavour Hadron Production at Fixed-Target Energies with STAR"*

Quark Matter 2023 September 2023
Houston, TX

Poster: *"Towards fixed-target light-flavor hadron production measurements with STAR from $\sqrt{s_{\mathrm{NN}}}$ = 3.2 – 7.7 GeV"*

California EIC Consortium Meeting January 2023
Los Angeles, CA (remote)

Talk: *"Temperature measurements of cooling foam apparatus"*

THESES

M.Sc. Thesis June 2022
"Measurements of flow vector fluctuations with ALICE"

[Thesis PDF]

B.Sc. Honours thesis June 2020
"Quantifying pulmonary fibrosis in murine histology"

EXPERIMENTAL OPERATIONS

STAR Shift Leader January 2026
Relativistic Heavy Ion Collider

Lead data taking responsibilities in the STAR control room. Multiple occasions, most recent date recorded.

STAR Detector Operator January 2026
Relativistic Heavy Ion Collider

Responsible for operation of the STAR detector. Multiple occasions, most recent date recorded.

ALICE Digital Control System Operator October 2021
Large Hadron Collider

Responsible for operation of the ALICE detector control systems.

SERVICE AND MENTORSHIP

Organizer, STAR Collaboration meeting at UC Davis

STAR Junior Representative

Organizer, Physics and Astronomy Peer Mentoring Program

STAR God Parent Committee member

Reviewer, Journal of Subatomic Particles and Cosmology

Reviewer, Physical Review Letters

STAR post-operations task force

Graduate Student Research Supervision

Undergraduate Nuclear Physics Research Mentor

UC Davis California EIC Consortium Meeting Co-Organizer

UC Davis Nuclear Physics Group Website Maintainer

UC Davis Nuclear Physics Computing Cluster Maintainer

TEACHING

Physics for the Biological Sciences Winter 2023
University of California, Davis

Introductory class in physics for undergraduates. Led group discussions, mandatory bi-weekly tutorials, and peer-based problem-solving sessions. Multiple occasions, most recent academic quarter listed.

Electronics Laboratory Fall 2024
University of California, Davis

Electronics and statistics laboratory for physics undergraduates. Taught basic circuit analysis and design, as well as basic data analysis techniques using Python. Multiple occasions, most recent academic quarter listed.

Advanced Electronics Laboratory Fall 2025
University of California, Davis

Active components and digital electronics. Multiple occasions, most recent academic quarter listed.

Anonymous student reviews available upon request.

SKILLS AND MEMBERSHIPS

Languages	C++, Python, Bash, LaTeX
Analysis frameworks	ROOT, NumPy/SciPy, Matplotlib
Simulation	Geant4, UrQMD, PHSD-PHQMD, SMASH, BeAGLE, Glauber Monte Carlo
Machine learning	scikit-learn, PyTorch, boosted decision trees
Computing	Slurm, HTCondor, Git/GitHub, Linux system administration
Memberships	American Physical Society, APS Division of Nuclear Physics